higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 3

(12030373)

11 March 2019 (X-Paper)
09:00–12:00

Calculators and drawing instruments may be used.

This question paper consists of 5 pages and 2 diagram sheets.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Sketches should be neat, labelled and large enough so that all details are clear.
6. Write neatly and legibly.

NCV

QUESTION 1: COLD-WATER RETICULATION

1.1 What is the purpose of a water meter? (1)

1.2 Where should water meters be installed? (1)

1.3 What is the static pressure in a cold-water system? (2)

1.4 Name FOUR possible constraints (limiting factors) when planning a route for the supply pipe from the water meter to the house. (4)

1.5 Describe, with the aid of a neat sketch, how siphonic action occurs in a tube that is the shape of an inverted (upside down) U-tube. (10)

1.6 What is the function of an in-line strainer? (2)
[20]

QUESTION 2: HOT-WATER RETICULATION

2.1 Give TWO disadvantages of a combination geyser. (2)

2.2 FIGURE 1, DIAGRAM SHEET 1 (attached), shows a diagrammatic layout of a typical push-through geyser.

Write the name of each component next to the number (1–10) in the ANSWER BOOK. (10 × 1) (10)

2.3 Give the colour codes for geyser pressure-relief valves with each of the following pressure ratings:

2.3.1 50 kPa

2.3.2 75 kPa

2.3.3 100 kPa

2.3.4 200 kPa

2.3.5 400 kPa

(5 × 1) (5)

2.4 Briefly describe the function of a pressure-relief valve when installed with a pressure-control valve in a closed (unvented) hot-water installation. (3)
[20]

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QUESTION 3: ABOVE-GROUND DRAINAGE SYSTEMS

3.1 Show, with the aid of a sketch, the difference between tubular P- and S-traps. Show the water seal, soffit and crown. (6)

3.2 Name THREE types of storage cisterns that supply water to a toilet. (3)

3.3 Explain the principle of trap venting. (3)

3.4 List THREE requirements for a good drainage system. (3)
[15]

QUESTION 4: SANITARY INSTALLATION

4.1 Write each of the following sanitary fittings and fixtures abbreviations in full:

4.1.1 AO

4.1.2 BT

4.1.3 CE

4.1.4 IE

4.1.5 WHB

(5 × 1) (5)

4.2 List FOUR ways to install or fit taps or a mixer to a basin. (4)

4.3 Why does a water closet NOT need to be fitted with a water seal trap? (1)

4.4 On which sanitary fitment or pipework is each of the following taps or valves used in the cold-water installations of a dwelling:

4.4.1 Pillar cock

4.4.2 Bibcock

4.4.3 Ball valve

4.4.4 Stopcock

4.4.5 Full-way valve

(5 × 1) (5)
[15]

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QUESTION 5: RAINWATER SYSTEM

5.1 Briefly explain how to measure, align and fix gutter brackets to a fascia at the correct gradient. (8)

5.2 What is the function of a rainwater system? (2)

5.3 Give THREE materials used to manufacture a gutter and rainwater downpipe. (3)

5.4 Name TWO methods of joining two sheet metals together. (2)

[15]

QUESTION 6: VENT-PIPE FLUSHING

FIGURE 2, DIAGRAM SHEET 2 (attached), shows the frustrum of a right cone where the bottom is NOT parallel to the base.

Draw, to scale 1:5, a full-pattern development of the frustrum of the right cone. **[10]**

QUESTION 7: TRENCHING, EXCAVATION AND TRENCH SUPPORT

Show, with a labelled sketch, what is meant by the term *invert depth*. **[5]**

TOTAL: 100

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DIAGRAM SHEET 1

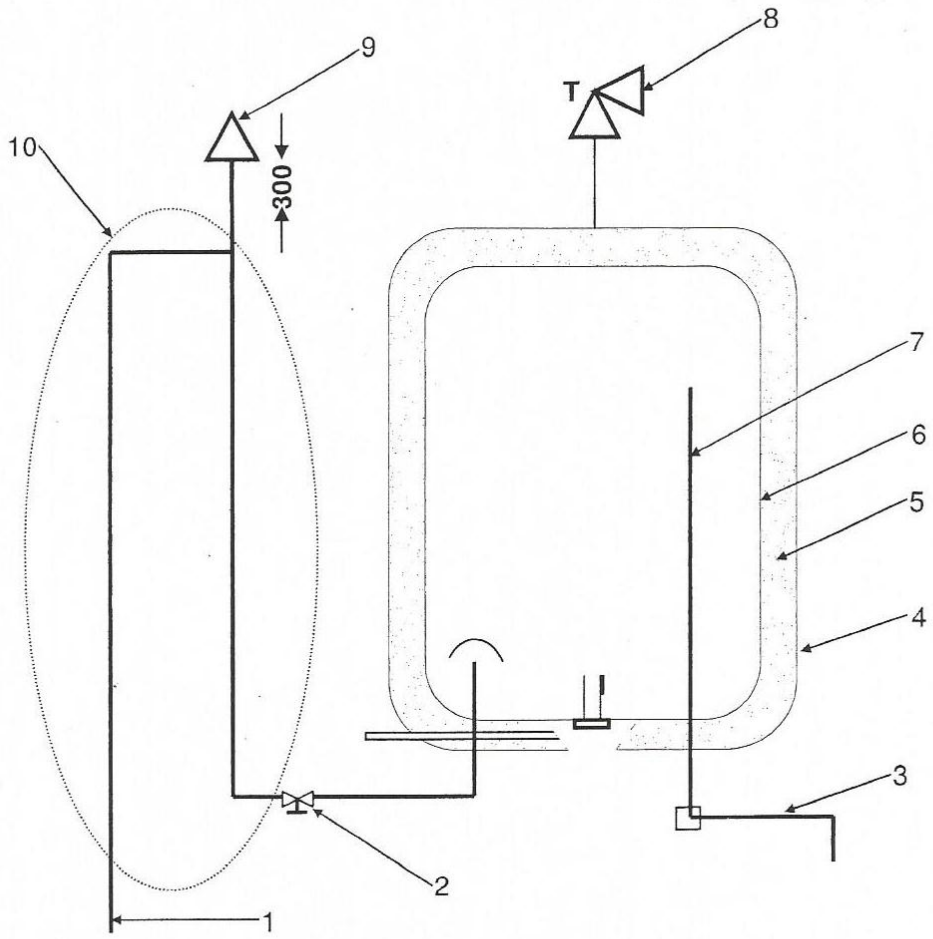


FIGURE 1

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DIAGRAM SHEET 2

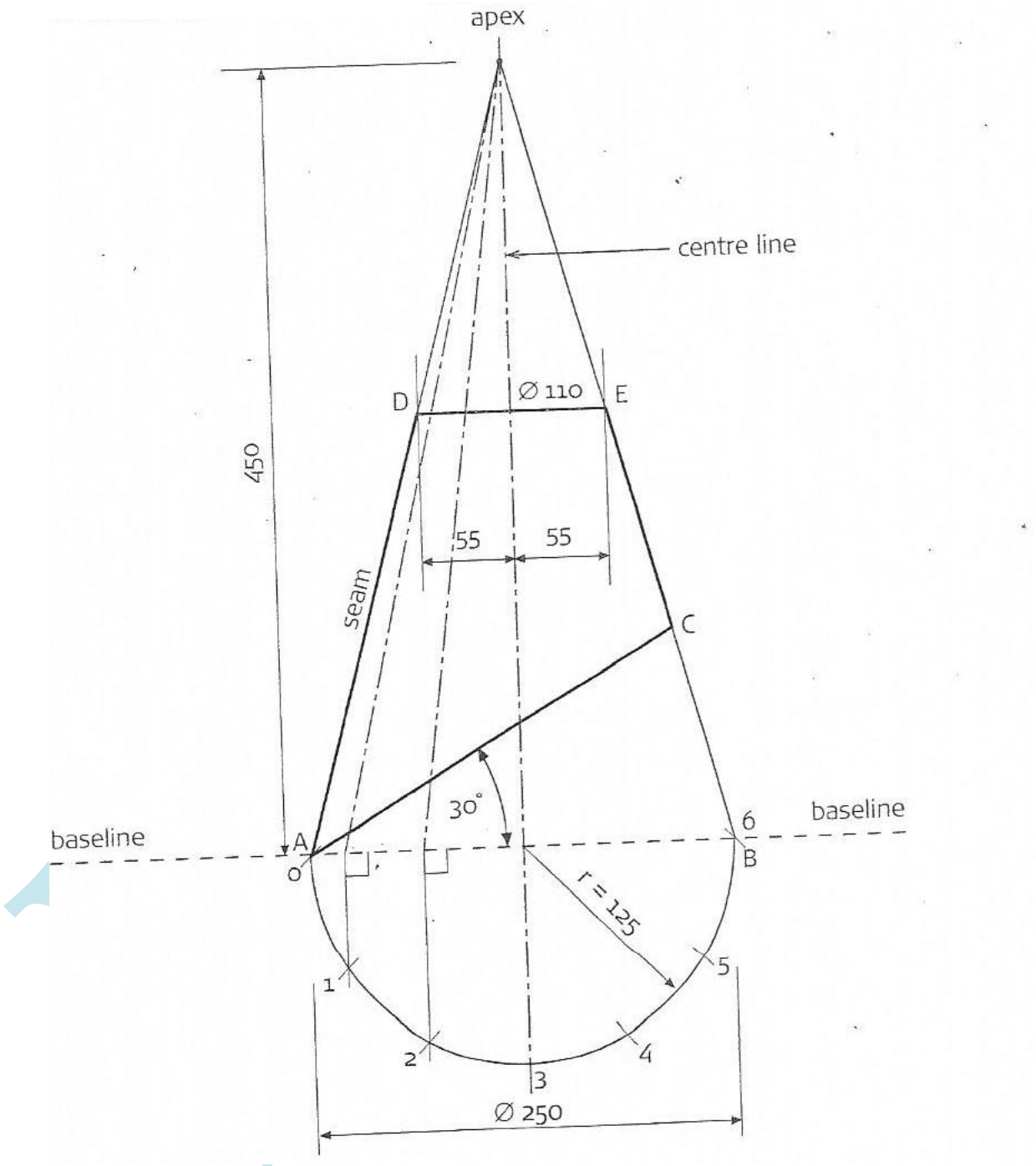


FIGURE 2

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